POTENTIAL IMPACTS OF GLOBAL CHANGES ON A BRAZILIAN CONTINENTAL SHELF AND SLOPE COMMUNITY

NASCIMENTO M. C. [1], OKEY, T. A. [2], VELASCO, G. [3]

[1] Post Graduation program in Ecology, Institute of Biology CP 6109, University of Campinas – UNICAMP 13083-970, Campinas, SP, Brazil. nascimento.marcelac@gmail.com.
[2] School of Environmental Studies, University of Victoria, PO Box 1700, STN CSC, Victoria, BC V8W 2Y2, Canada; Thomas.Okey@gmail.com:
[3] Instituto de Oceanografia, Universidade Federal do Rio Grande, Rio Grande, RS, Brazil; P.O. Box 474. gonzalo.velasco@furg.br

INTRODUTION

Global climate change is affecting biological communities of continental shelves and slopes. The intensity of effects on each species, population, and individual is mediated by indirect effects via biological relationships in addition to sensitivities and tolerances to direct effects. Because each species has unique environmental tolerances, the effects of the local manifestations of global changes will be species-specific and locally–specific, and will lead to re-shuffling of species and changes in ecosystem-function.

MATERIAL AND METHODS

In this pilot exploration, we examined the effects of projected global changes in primary production on a Brazilian continental shelf food web model constructed by Nascimento et al. (2012 – Figure 1) given three scenarios of global temperature change. At this stage, these explorations were conducted without knowledge of local and spatial heterogeneity and without explicit knowledge of tolerances and sensitivities of particular functional groups other than how changes in primary production cascades through the present model tropically.

We examined three scenarios of changes in primary production, two of them extreme and one intermediate (Princeton/Marra et al. (2003) (1.096), Princeton/Poly (0.989) and Princeton/Dunne (1.046). For each value, we made the simulations using the global changes scenario of lower (1.4°C) high (5.8°C) and intermediate (3.6°C) temperature increases. We ran the simulations for 50 years with and without the global change scenarios, while maintaining the current level of fisheries

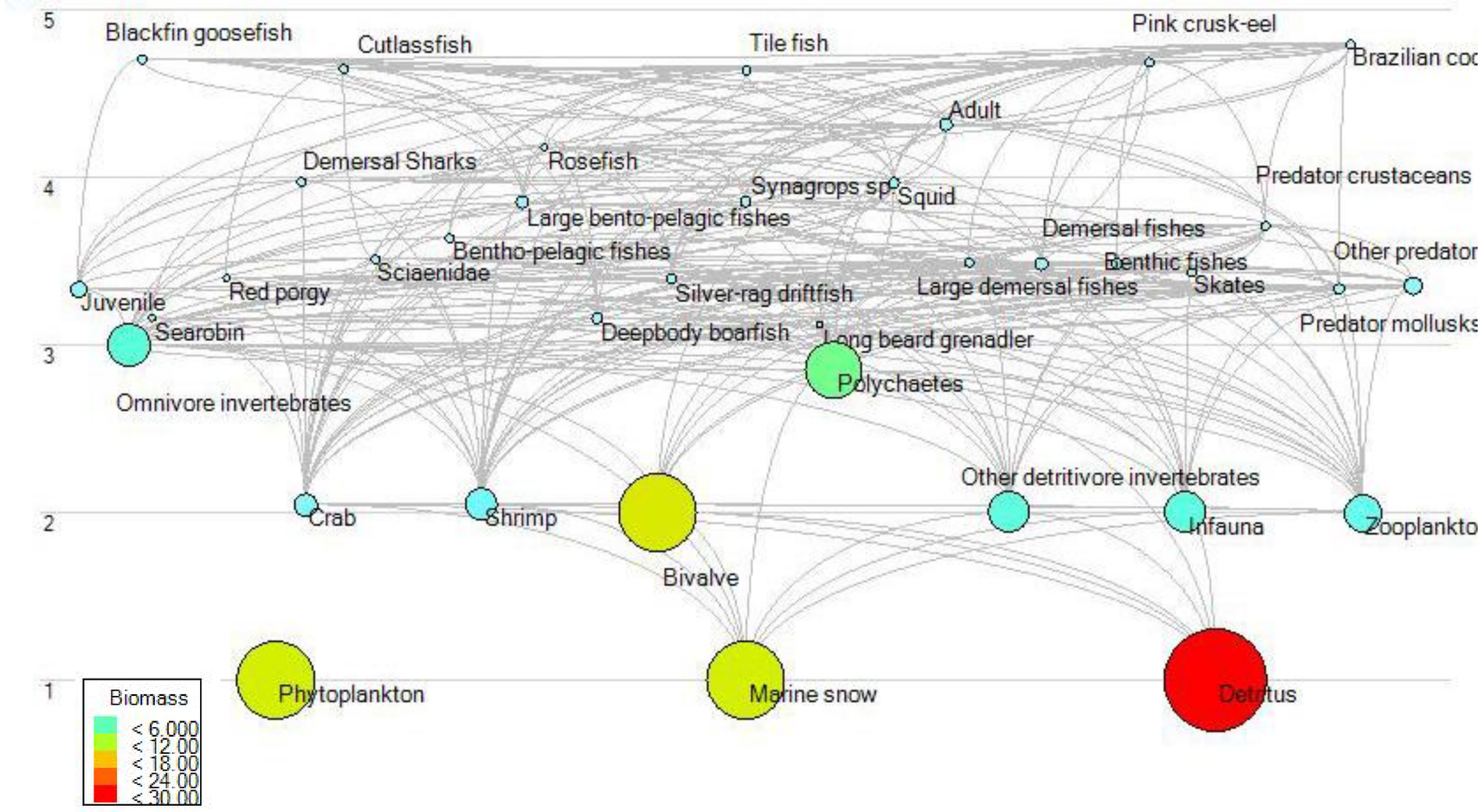


Fig 1. The initial model - Nascimento *et al.* 2012

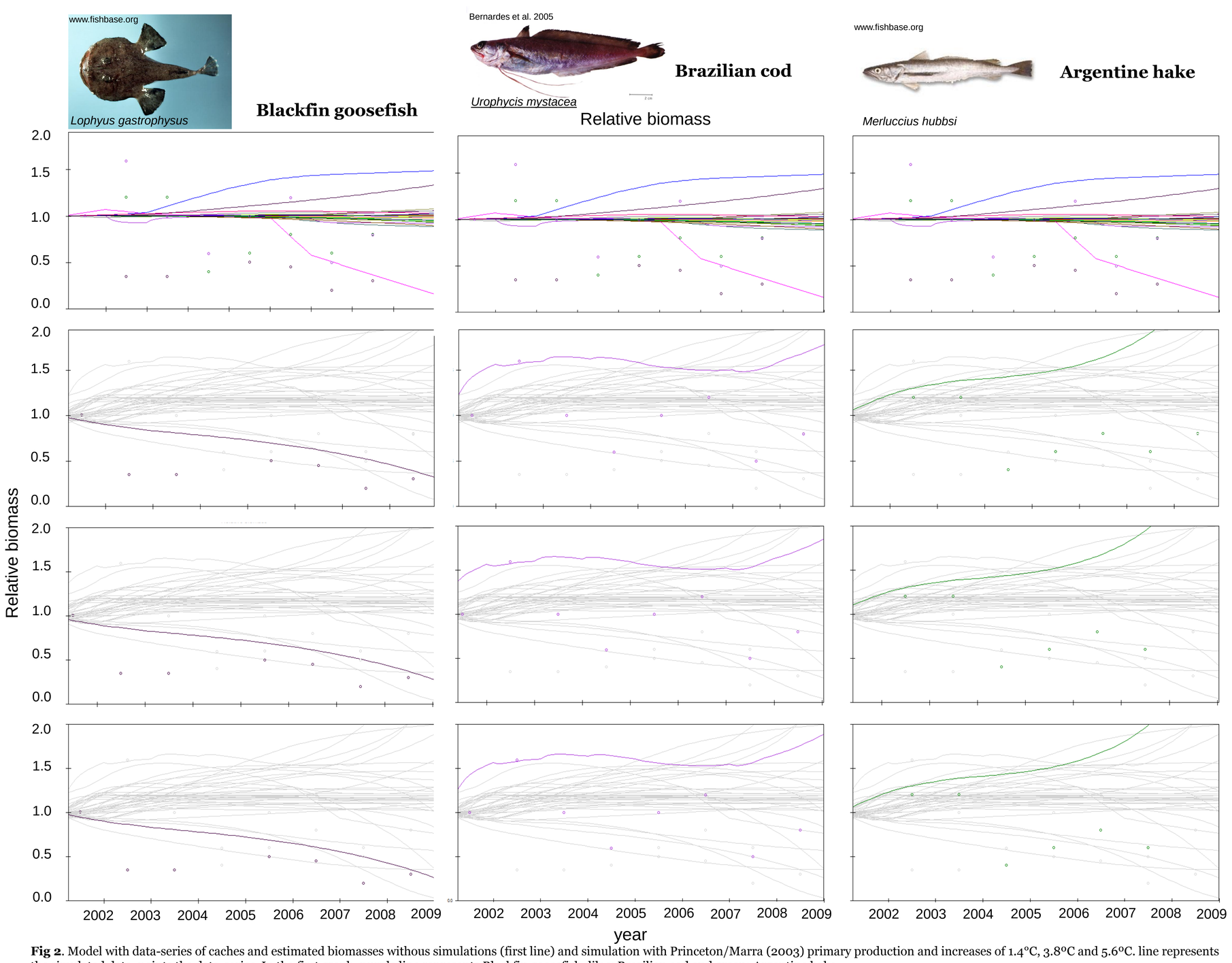


Fig 2. Model with data-series of caches and estimated biomasses withous simulations (first line) and simulation with Princeton/Marra (2003) primary production and increases of 1.4°C, 3.8°C and 5.6°C. line represents the simulated data, points the data-series. In the first graph, purple line represents Blackfin goosefish, lilac: Brazilian cod and green: Argentine hake.

RESULTS

The simulations without global change indicated high increases in the initial biomass (2001) of five functional groups over the course of 50 years. Also we observed the collapse of 'Large bentho-pelagic fishes' in 2039, and the strong decrease of the biomass of several functional groups, which was attributed to fisheries pressure in Nascimento et al. (2012).

The simulations that specified global changes in primary production also indicated increases in the biomasses of the same functional groups, but at different magnitudes, depending of the scenario.

For some species, such as the Blackfin goosefish, accounting for global changes in productivity lead to the best fit of the simulated trajectory with the observed trajectory over the last 10 years, especially the simulation with the largest increase in primary production.

ACKNOWLEDGMENTS

Villy Christensen, Marta Coll and Cecilia Amaral for their technical help, CAPES and CNPq commissions, from Brazil, that supplied the first author with scholarships. TAO thanks the Pew Fellows Program in Marine Conservation, Pew Environment Group